

E1

Consider the circuit in Fig. 1. The circuit parameters at  $T = 25^\circ\text{C}$  are the following:

$V_{CC} = 10\text{ V}$ ;  $R_{B1} = 1.8\text{ M}\Omega$ ;  $R_{B2} = 1.5\text{ M}\Omega$ ;  $R_{E1} = 0.47\text{ k}\Omega$ ;  $R_{C1} = 1\text{ k}\Omega$ ;  $R_{C2} = 4.7\text{ k}\Omega$ ;  $R_F = 22\text{ k}\Omega$ ;  $R_s = 1\text{ k}\Omega$ ;  $C_M = 10\text{ pF}$ .

$Q_1$  (PNP BJT):  $V_{BE1} = -0.7\text{ V}$ ;  $\beta_{F1} = \beta_{01} = 150$ ;  $V_{CEsat} = -0.2\text{ V}$ ;  $r_0 = +\infty$ ;

$Q_2$  (NPN BJT):  $V_{BE2} = 0.7\text{ V}$ ;  $\beta_{F2} = \beta_{02} = 200$ ;  $V_{CEsat} = 0.2\text{ V}$ ;  $r_0 = +\infty$ ;

$V_T = 26\text{ mV}$ .

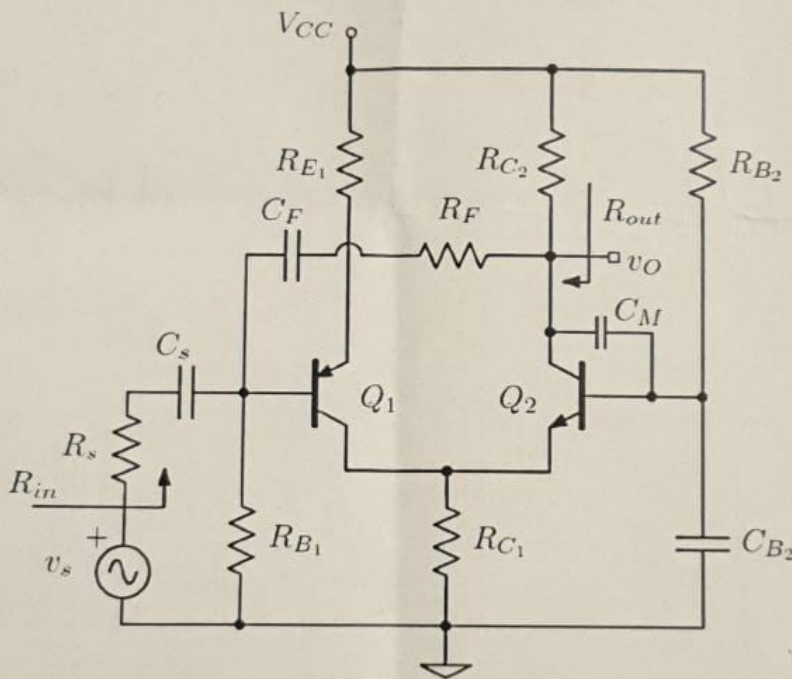


Fig. 1 – A case of two-stage feedback amplifier.

Considering *all capacitors* to be equivalent to *open circuits*, determine:

- the operating points ( $V_{CE}$ ,  $I_C$ ) of  $Q_1$  and  $Q_2$ .

$Q_1 (-7.8\text{ V}, 0.745\text{ mA})$   $Q_2 (3.51\text{ V}, 1.059\text{ mA})$

Assuming that, at the frequencies of interest (i.e. at mid-band),  $C_s$ ,  $C_F$  and  $C_{B2}$  can be considered equivalent to *short circuits*, while  $C_M$  can be considered equivalent to an open circuit, determine also:

- the voltage gain  $A_v = v_o/v_s$  of the amplifier;
- the input resistance  $R_{in}$  indicated in the figure;

$-5.36$   
 $3.5\text{ k}$

4. the output resistance  $R_{out}$  indicated in the figure.

2.93k

Finally, considering only the effect of capacitor  $C_M$ :

5. the high frequency bandwidth limit  $f_H$  of the amplifier.

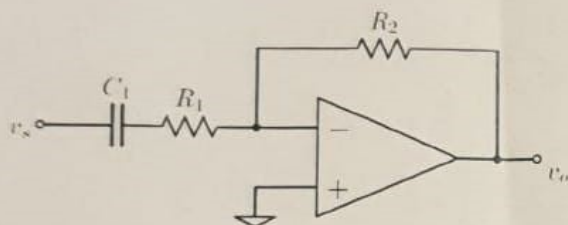
5.44M

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_2 = 22 \text{ k}\Omega$ ;  $C_1 = 100 \text{ nF}$ ;

OpAmp differential gain:  $A_{d0} = 10^5 \text{ [V/V]}$ ,  $\omega_0 = 10 \text{ rad/s}$ .



$$A_d(s) = \frac{A_{d0}}{1 + \frac{s}{\omega_0}}$$

Fig. 2 – Compensated OpAmp in the inverting configuration.

Considering initially  $R_1 = 0 \Omega$ , determine:

1. the expression of the *ideal* circuit voltage gain  $A_v(s) = v_o/v_s$ ;  $-R_2/SC_1$
2. an estimation of the circuit *phase margin*;  $1.25$
3. the *block diagram* of the amplifier, according to *feedback theory*;
4. the expression of the *input scaling function*  $k_s(s)$  in the above diagram.  $SC_1$

Then, determine

5. the value of resistor  $R_1$  that makes the phase margin of the circuit equal to  $60^\circ$ .

117.37